

Lecture 3: Convolutional Neural Networks
Regularisation, Class Activation Maps
Adversarial Examples and Limits

Deep Learning for Scientific Discovery

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Recap: MNIST, the Deep Learning "Hello World"

1 output, MSE

$\mathcal{L}_{\text{MSE}} = (\bar{y} - y)^2$

$y \in \{0, \ldots, 9\}$: true digit, $\bar{y}$: network output

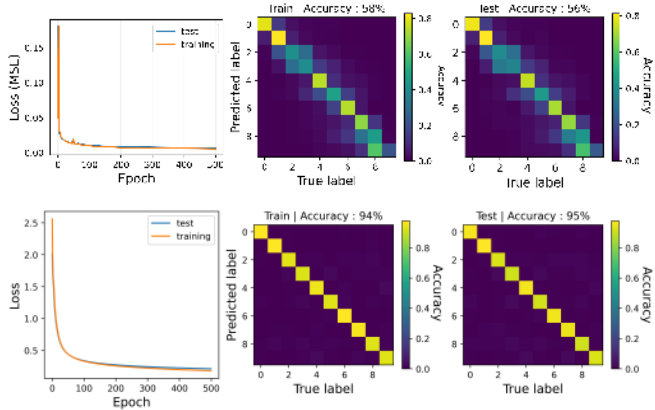
test accuracy 56%

10 outputs, cross-entropy

$\mathcal{L}_{\text{CE}} = -\log q(\text{true digit})$

$q(k)$: softmax output for digit k

test accuracy 95%



Losses per image, averaged over the training set; same hidden layer (21 sigmoid nodes). Confusion matrix: row = predicted, column = true digit. A single output orders the digits, so MSE errors lie next to the diagonal (3 taken for 2, 9 for 8).

the formulation of the problem matters as much as the network

1

Hello World #2: CIFAR-10 and Convolutional Networks

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CIFAR-10 Dataset



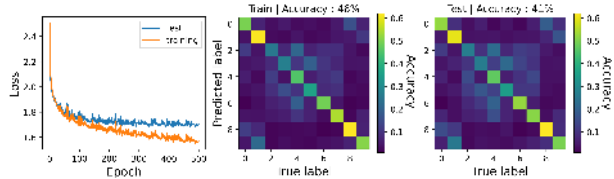
- 10 classes: aeroplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck
- 50 000 training, 10 000 test images
- 32×32 pixels in **3 colour channels** (R, G, B): each image is a $32 \times 32 \times 3$ tensor
- MNIST was $28 \times 28 \times 1$

A. Krizhevsky, *Learning Multiple Layers of Features from Tiny Images* (2009).

CIFAR-10: The Same Naive Network + Cross-Entropy

Exactly the MNIST recipe:

- **flatten** the image:
  $32 \cdot 32 \cdot 3 = 3072$ inputs
- one hidden layer, 21 sigmoid nodes
- 10 outputs, softmax, cross-entropy



$3072 \cdot 21 + 21 + 21 \cdot 10 + 10 = 64\,753$

parameters

- **Poor performance:** about 46% training, 41% test accuracy — the model *underfits*

What do we lack in our approach?

Flattening Loses Spatial Structure

Neighbouring pixels are correlated: an edge, a texture, an eye are *local* patterns of a few nearby pixels.

After flattening, the dense layer treats pixel #1 and pixel #2 as unrelated as pixel #1 and pixel #3000.

Shuffle all pixels with one fixed permutation: a dense network learns equally well — it never used the 2D layout.

What an image model should know

- **locality:** look at small neighbourhoods
- **weight sharing:** the same edge detector is useful everywhere in the image
- **translation:** a cat in the corner is still a cat

$\Rightarrow$ **capture spatial correlations** with *convolutions*.

Convolution: the Continuous Definition (1D)

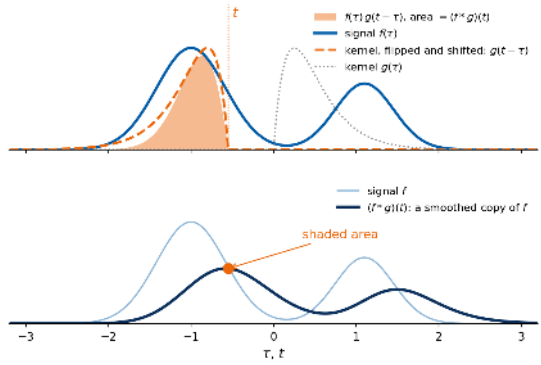
Two functions $f(\tau)$ (the signal) and $g(\tau)$ (the kernel):

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)\, g(t - \tau)\, d\tau$$

For each output point t:

- **flip** the kernel, $g(-\tau)$
- **shift** it by t, $g(t - \tau)$
- **multiply** with $f(\tau)$
- **integrate**: the shaded area is $(f * g)(t)$

Moving t traces the whole curve $f * g$, here a **smoothed** copy of f.



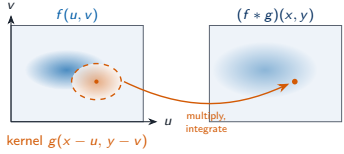
Convolution: the Continuous Definition (2D)

For functions of two variables, e.g. an image $f(x, y)$ and a kernel $g(x, y)$:

$$(f * g)(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u, v)\, g(x - u,\, y - v)\, du\, dv$$

the same operations as in 1D: flip g in both directions, shift it to (x, y), multiply with f, integrate over the plane; one number for each point (x, y)

g a 2D Gaussian $\Rightarrow$ $f * g$ is f **blurred**: every point replaced by a weighted average of its neighbourhood



the kernel, centred at (x, y), weights the image around that point; the integral gives the output there

Convolution theorem: in Fourier space a convolution is a product

$$\mathcal{F}[f](k) = \int f(t)\, e^{-ikt}\, dt, \qquad \mathcal{F}[f * g](k) = \mathcal{F}[f](k) \cdot \mathcal{F}[g](k), \qquad f * g = \mathcal{F}^{-1}\big[\mathcal{F}[f] \cdot \mathcal{F}[g]\big]$$

in 2D the same, with $e^{-i(k_x x + k_y y)}$

Also: linear; commutative, $f * g = g * f$; shift-equivariant, $f(\cdot - a) * g = (f * g)(\cdot - a)$.

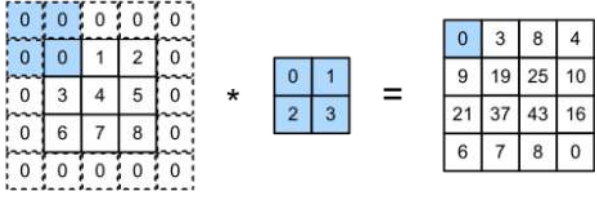
Discrete Convolution in 2D

Image $x(i,j)$, kernel $G(m,n)$ of size $k \times k$:

$$\text{convolution:} \quad (G * x)(i,j) = \sum_m \sum_n G(m,n)\, x(i-m,\, j-n)$$

$$\text{cross-correlation (what CNN layers compute):} \quad S(i,j) = \sum_m \sum_n G(m,n)\, x(i+m,\, j+n)$$

The two differ only by flipping the kernel. The kernel is *learned*, so the flip does not matter: deep-learning libraries compute the cross-correlation and still call it "convolution".

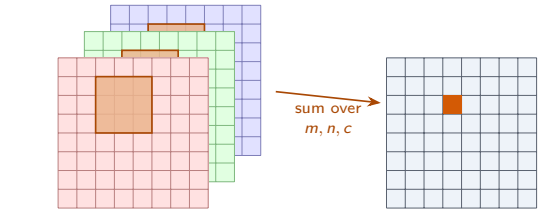


a 2×2 kernel slides over a zero-padded 3×3 input; each output is the sum of element-wise products, e.g.

$$0 \cdot 0 + 0 \cdot 1 + 0 \cdot 2 + 1 \cdot 3 = 3$$

Images with Three Channels

A colour image has three numbers per pixel, red, green and blue: $x(i, j, c)$ with $c \in \{R, G, B\}$. A CIFAR-10 image is $32 \times 32 \times 3$.



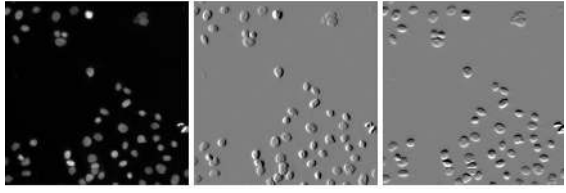
$$S(i, j \mid c) = \sum_{m,n} G(m, n, c)\, x(i + m,\, j + n,\, c), \qquad S(i, j) = \sum_{c} S(i, j \mid c) + b$$

each channel is convolved with its own slice of the kernel, and the three results are added: one kernel gives *one* feature map; C_{out} kernels give C_{out} feature maps, which are the channels of the next layer's input

Kernels Detect Features: Sobel Filters

$$\vec{G}_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \qquad \vec{G}_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

approximate the intensity gradient: they light up where the image changes sharply — **edges**



human cell nuclei, fluorescence microscope · vertical edges ($\vec{G}_x$) · horizontal edges ($\vec{G}_y$)

Sharpening, blurring and the identity are 3×3 kernels too.

Idea of a CNN: do not design the kernels, learn them.

image: D. Root, CC0, from J. Moffat et al., *A Lentiviral RNAi Library for Human and Mouse Genes Applied to an Arrayed Viral High-Content Screen*, Cell **124**, 1283 (2006), doi:10.1016/j.cell.2006.01.040

Kernels Are Learnable Parameters

-1	0	$+1$
-2	0	$+2$
-1	0	$+1$

Sobel $\vec{G}_x$:
fixed, designed by hand

G_{11}	G_{12}	G_{13}
G_{21}	G_{22}	G_{23}
G_{31}	G_{32}	G_{33}

CNN kernel:
9 trainable weights

- the entries $G(m, n, c)$ and the bias b are the **weights** of a convolutional layer, as W and b are for a dense layer
- they start random and are trained by gradient descent on the same loss:

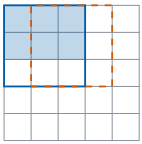
$$G(m, n, c) \leftarrow G(m, n, c) - \eta \frac{\partial \mathcal{L}}{\partial G(m, n, c)}$$

- backpropagation gives the gradient; one kernel is shared by all positions, so its gradient sums over all of them

After training, first-layer kernels often look like edge detectors.

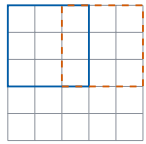
Kernel Size, Stride, Padding

stride 1



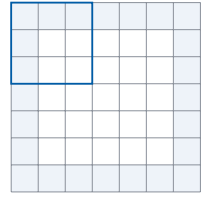
5×5 input, 3×3 kernel $\rightarrow 3 \times 3$ output

stride 2



the kernel jumps 2 pixels $\rightarrow 2 \times 2$ output

padding 1



zeros around the border $\rightarrow 5 \times 5$ output ("same")

Output size along one dimension

$$O = \left\lfloor \frac{W - k + 2P}{s} \right\rfloor + 1$$

W : input size, k : kernel size
P : padding, s : stride

CIFAR-10: $W = 32$, $k = 3$, $P = 1$, $s = 1 \Rightarrow O = 32$; with $s = 2$: $O = 16$

Pooling: Max, Average, Global Average

Window 2×2, stride 2: each 2×2 block of the feature map becomes *one* number.

$$\begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 7 & 8 \\ 4 & 2 & 1 & 3 \\ 9 & 8 & 6 & 5 \end{bmatrix}$$

$$\xrightarrow{\text{max pooling}} \begin{bmatrix} 6 & 8 \\ 9 & 6 \end{bmatrix}$$

$$\xrightarrow{\text{average pooling}} \begin{bmatrix} 3.75 & 5.25 \\ 5.75 & 3.75 \end{bmatrix}$$

$$\xrightarrow{\text{global average pooling}} 4.625$$

top-left block: $\max(1,3,5,6) = 6$, $\quad (1+3+5+6)/4 = 3.75$; $\quad$ global: $74/16 = 4.625$

- **max pooling**: the strongest response; used between conv layers
- **global average pooling**: one number per feature map; used at the end of a network
- no trainable parameters

Downsampling

4 times fewer numbers; the next 3×3 kernel covers a 6×6 region; small shifts do not change the output. The price: exact positions are lost.

CNNs Learn a Hierarchy of Features

- a CNN is a neural network that **learns the kernels** for each channel
- layers of kernels capture a **hierarchy of features**: edges $\rightarrow$ parts $\rightarrow$ objects



first layer | middle layer | deep layer

features learned from face images by a convolutional deep belief network: H. Lee, R. Grosse, R. Ranganath, A. Y. Ng, *Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations*, ICML 2009, doi:10.1145/1553374.1553453

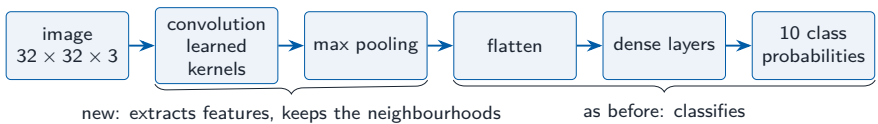
Second Attempt at CIFAR-10 Classification

First attempt: a dense network

- the image is flattened into 3072 numbers
- which pixels are neighbours is thrown away
- 46% training, 41% test accuracy

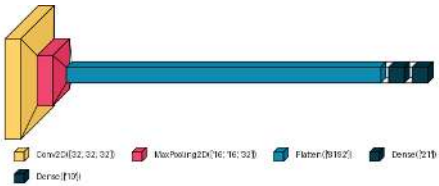
Second attempt: a convolutional network

- the image stays an image: $32 \times 32 \times 3$
- kernels look at small neighbourhoods and are *learned*
- pooling reduces the size of the feature maps

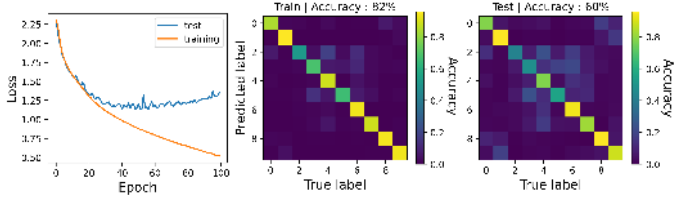


Same data, same loss (categorical cross-entropy), same training loop: **only the model changes.**

A Small CNN on CIFAR-10

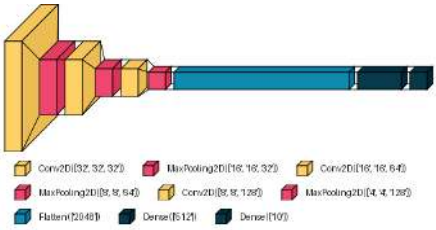


- Conv2D: 32 kernels 3×3, sigmoid, padding "same" $\rightarrow 32 \times 32 \times 32$
- MaxPooling $2 \times 2 \rightarrow 16 \times 16 \times 32$
- flatten $\rightarrow$ 8192, dense 21 (sigmoid), dense 10 (softmax)

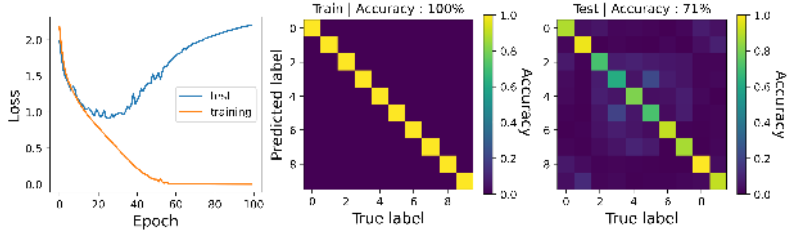


About 82% training, 60% test accuracy — better than 41%, but the test loss **rises after ∼40 epochs**: *overfitting*.

More CNN Layers: Still Overfitting



three blocks Conv2D (ReLU) + MaxPooling:
$32 \times 32 \times 32 \to 16 \times 16 \times 64 \to 8 \times 8 \times 128$,
then flatten, dense 512, dense 10



100% training, 71% test accuracy: **more complexity does not cure overfitting**.

2

Regularisation: Curing Overfitting

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Regularisation: Three Common Techniques

Regularisation: any technique that constrains training or model capacity so that the network *generalises* to unseen data instead of memorising the training set.

L1 / L2 penalty	Dropout	Batch normalisation
a term in the loss that keeps the weights small	randomly switch off neurons during training	normalise the inputs of each hidden layer over a mini-batch

Also widely used: *early stopping* (stop when the validation loss starts to rise), *data augmentation* (shifted, flipped, cropped copies of the training images), *learning-rate scheduling*.

L1 / L2 Regularisation

L1 (Lasso)

$$\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{original}} + \lambda \sum_i |\theta_i|$$

drives some weights exactly to zero: *sparse* solutions

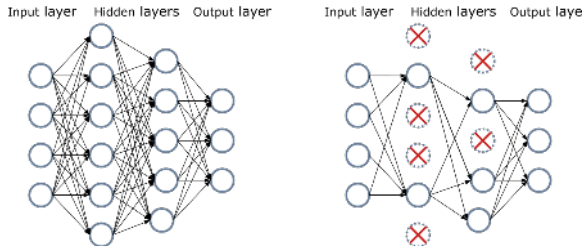
L2 (Ridge, "weight decay")

$$\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{original}} + \lambda \sum_i \theta_i^2$$

discourages large weights: *smoother* models

- θ_i: the weights of the network; λ: the regularisation strength, a hyperparameter
- a **soft constraint**: weights are encouraged, not forced, to stay small — λ plays the role of a Lagrange multiplier
- $\lambda = 0$: no regularisation; λ too large: the network underfits

Dropout



- at every training step each neuron of the layer is switched off with probability p (the *dropout rate*, typically 0.2–0.5)
- the network cannot rely on any single neuron: it learns **redundant representations**
- each step trains a different sub-network: dropout acts as an **ensemble** of sub-networks
- at test time all neurons are active

N. Srivastava et al., *Dropout: A Simple Way to Prevent Neural Networks from Overfitting*, JMLR **15**, 1929 (2014), jmlr.org/papers/v15/srivastava14a.html

Batch Normalisation

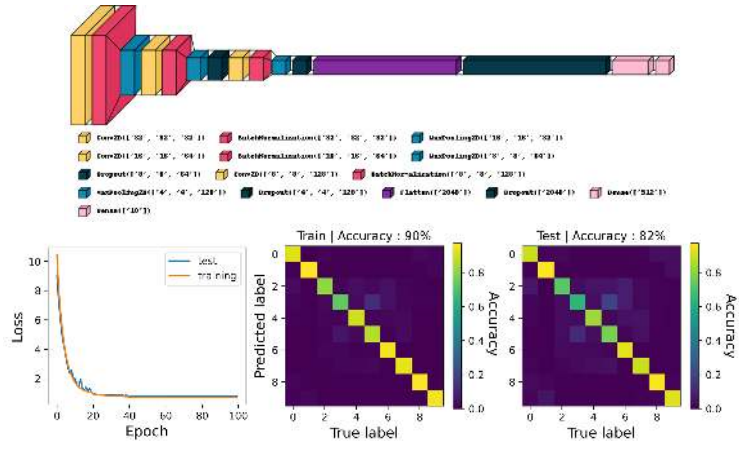
A BatchNorm layer normalises the input of a hidden layer over the mini-batch $\{\mathbf{x}_j\}_{j=1}^{N_B}$:

$$\mu_B = \frac{1}{N_B}\sum_{j=1}^{N_B}\mathbf{x}_j, \qquad \sigma_B^2 = \frac{1}{N_B}\sum_{j=1}^{N_B}(\mathbf{x}_j-\mu_B)^2 \qquad \mathbf{x}_j \leftarrow \frac{\mathbf{x}_j-\mu_B}{\sqrt{\sigma_B^2+\epsilon}}, \qquad \mathbf{x}_j \leftarrow \gamma\,\mathbf{x}_j+\beta$$

- normalise: zero mean and unit variance within the mini-batch (ϵ: a small constant for numerical stability)
- then rescale and shift with **trainable** γ and β
- makes the network less dependent on weight initialisation
- reduces sensitivity to weight changes; helps against exploding and vanishing gradients

S. Ioffe, C. Szegedy, *Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift*, ICML 2015, arXiv:1502.03167

The Larger CNN with Regularisation



Same three Conv2D blocks, now with batch normalisation and dropout: 90% training, 82% test accuracy.

Before: 100% / 71%. **The training and test losses now stay together.**

3

Class Activation Maps: Where the CNN Looks

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Class Activation Maps

input (32 x 32)



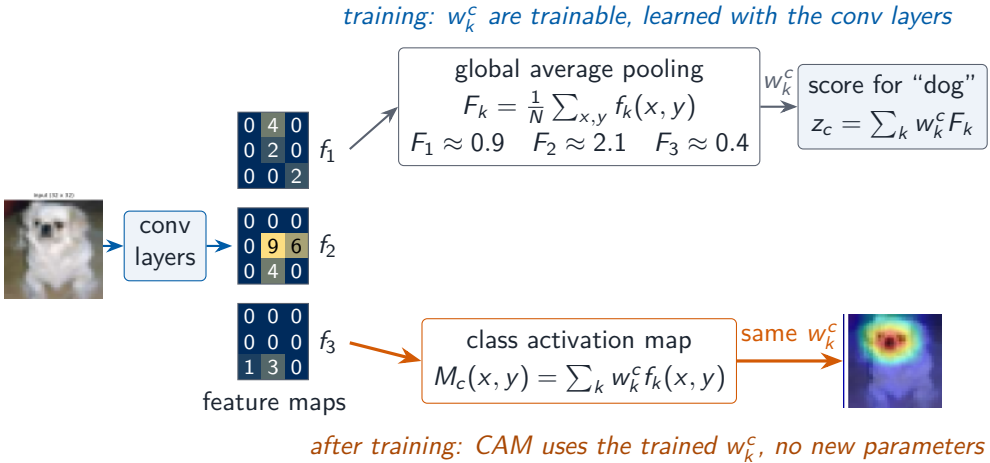
a CIFAR-10 test image; our CNN: "dog", 1.00

Which parts of the image were important for the prediction?

Can we get insight into what the CNN was looking at when it made its prediction?

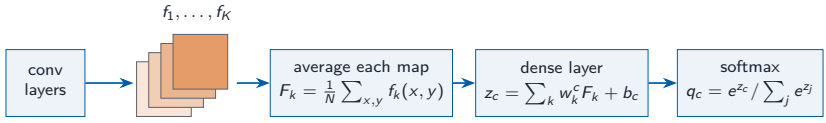
A **class activation map** (CAM) answers with a picture: a map over the image that shows *where* the evidence for a class is.

From Feature Maps to a Class Map



B. Zhou et al., *Learning Deep Features for Discriminative Localization*, CVPR 2016, arXiv:1512.04150

CAM: From the Class Score Back to a Map



K feature maps, each $h \times w$, $N = hw$ pixels

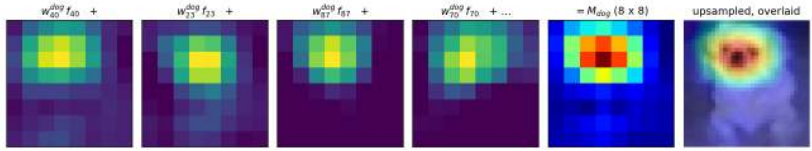
$$z_c = \sum_k w_k^c F_k + b_c = \sum_k w_k^c \frac{1}{N} \sum_{x,y} f_k(x,y) + b_c = \frac{1}{N} \sum_{x,y} \underbrace{\sum_k w_k^c f_k(x,y)}_{M_c(x,y)} + b_c$$

Class activation map

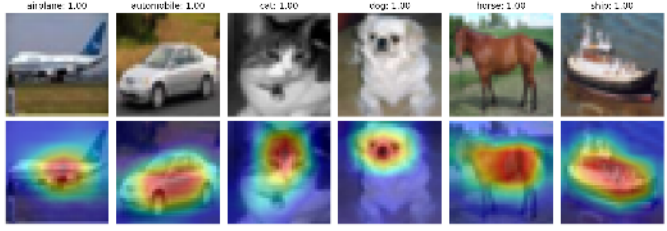
$$M_c(x,y) = \sum_k w_k^c f_k(x,y)$$

the class score z_c is the average of M_c over all positions, plus b_c

CAM on CIFAR-10



feature maps times their weights for "dog", added to one 8×8 map, enlarged and laid over the image



correctly classified test images: the evidence sits on the object, not on the background

our CNN with GAP: 6 conv layers $\rightarrow$ 128 maps of 8×8 $\rightarrow$ GAP $\rightarrow$ 10 outputs; 87.6% test accuracy

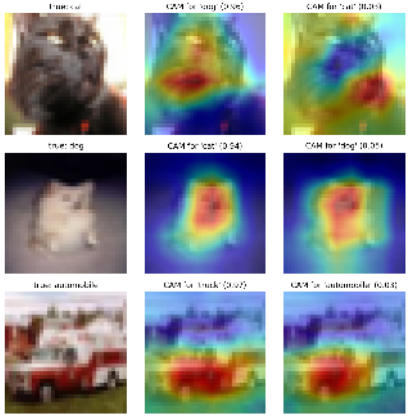
CAM for Two Classes in One Photograph



- the map of a dog class lies on the dog; the map of a cat class lies on the cat
- network: ResNet-18 trained on ImageNet (1000 classes), also ending with global average pooling
- p: softmax probability of that class. It is small: two small animals, 1000 classes to share the probability.

photograph: Socks and Buddy, U.S. National Archives, public domain, Wikimedia Commons

CAM: Mistakes and Limits



map of the predicted class (middle) and of the true class (right)

- **cat → "dog" (0.96):** a dark close-up; the "dog" evidence on the lower part of the face
- **dog → "cat"**, **automobile → "truck"**: both maps on the object, the wrong one of two similar classes

CAM tells us

where the evidence for a class is: on the object or on the background (shortcut learning)

CAM does not tell us

why that region mattered; fine detail (resolution of the last feature map, 8×8 here)

Accurate for the wrong reason: wolves told from huskies by the *snow* in the background. M. T. Ribeiro et al., *"Why Should I Trust You?": Explaining the Predictions of Any Classifier*, KDD 2016, arXiv:1602.04938

Grad-CAM: Class Activation Maps for Any CNN

- $f_k(x, y)$: feature maps of the last convolutional layer; z_c: score of class c before softmax
- CAM needs one weight w_k^c per map; when other layers follow the last convolution, Grad-CAM uses the gradient averaged over the N pixels: $\alpha_k^c = \dfrac{1}{N} \sum_{x,y} \dfrac{\partial z_c}{\partial f_k(x, y)}$

CAM

$$M_c(x, y) = \sum_k w_k^c \, f_k(x, y)$$

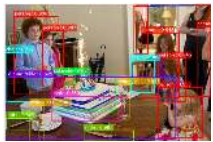
Grad-CAM

$$M_c(x, y) = \mathrm{ReLU}\Big(\sum_k \alpha_k^c \, f_k(x, y) \Big)$$

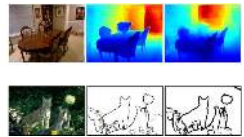
- ReLU keeps only the regions that *increase* z_c
- GAP network: $\alpha_k^c = w_k^c / N$, and Grad-CAM gives the CAM map

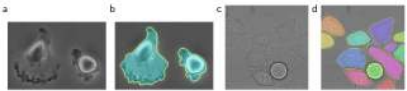
R. R. Selvaraju et al., *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*, ICCV 2017, arXiv:1610.02391

Detection and Segmentation

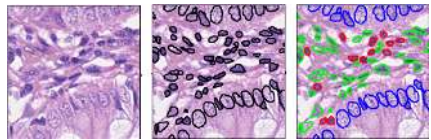








living cells: input (a, c), one colour per cell (b, d)



tissue: input, nuclei outlined, coloured by type

- **detection:** a box and a class for every object
- **segmentation:** a class for every pixel, and in cells and tissues every object separately: counts, sizes, shapes

O. Ronneberger, P. Fischer, T. Brox, *U-Net: Convolutional Networks for Biomedical Image Segmentation*, MICCAI 2015, arXiv:1505.04597; S. Graham et al., *HoVer-Net: Simultaneous Segmentation and Classification of Nuclei in Multi-Tissue Histology Images*, Med. Image Anal. **58**, 101563 (2019), doi:10.1016/j.media.2019.101563

Where We Are

data	model	train	test
MNIST	dense, 1 output, MSE	58%	56%
MNIST	dense, 10 outputs, cross-entropy	94%	95%
CIFAR-10	dense, 10 outputs, cross-entropy	46%	41%
CIFAR-10	small CNN	82%	60%
CIFAR-10	larger CNN	100%	71%
CIFAR-10	larger CNN with batch norm, dropout	90%	82%
CIFAR-10	CNN with GAP, batch norm, augmentation	91%	88%

first six rows: book Ch. 2; last row: our CAM network

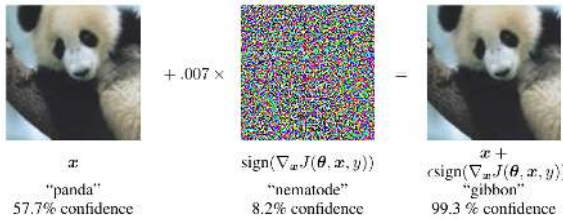
Regularisation closes the gap between training and test.

4

When Networks Fail: Adversarial Examples

JAGIELLONIAN UNIVERSITY
IN KRAKÓW

Adversarial Examples



- **left:** a photograph. The trained classifier says "panda", with 57.7% confidence — correct.
- **middle:** a pattern computed from the network. Shown alone it means nothing: the classifier's best guess is "nematode" (a roundworm), with 8.2%.
- **right:** photograph + 0.007× pattern. For us the same photograph; the classifier says "gibbon", with 99.3% confidence.

Such inputs are called **adversarial examples**.

I. J. Goodfellow, J. Shlens, C. Szegedy, *Explaining and Harnessing Adversarial Examples*, ICLR 2015, arXiv:1412.6572

One-Pixel Attack

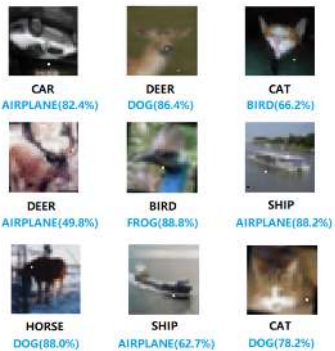


Fig. 1: CIFAR-10



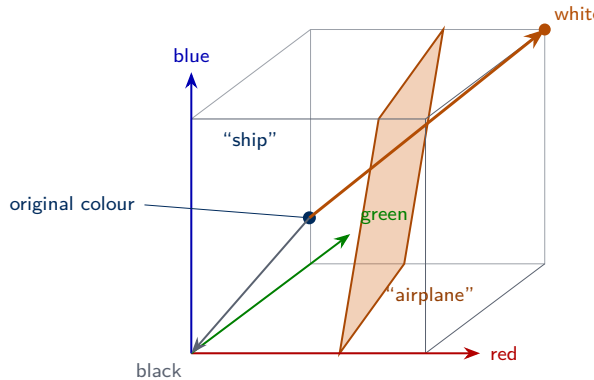
Fig. 2: ImageNet, pixel in the red circle

- black: the true class; blue: the class after changing one pixel, with the network's confidence
- Kaggle CIFAR-10 test set: 67.97% of the images can be moved to at least one other class by one pixel, with 74.03% confidence on average
- ImageNet: 16.04% of the test images, 22.91% confidence

J. Su, D. V. Vargas, K. Sakurai, *One Pixel Attack for Fooling Deep Neural Networks*, IEEE Trans. Evol. Comput. **23**, 828 (2019), doi:10.1109/TEVC.2019.2890858, arXiv:1710.08864

One-Pixel Attack: Mechanism

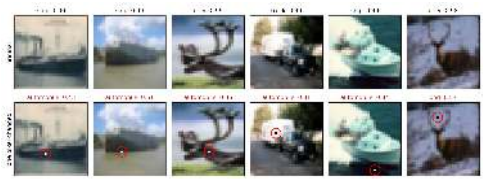
- an image is one point in a space with 3072 axes
- the network cuts this space into class regions, separated by **decision boundaries**
- one pixel = 3 of these axes: its red, green and blue value
- if a boundary is close along these 3 axes, changing the colour of that pixel moves the image into another class



the cube of all colours of *one* pixel; orange: the decision boundary. To white: the class changes. To black: it does not.

J. Su, D. V. Vargas, K. Sakurai, *One Pixel Attack for Fooling Deep Neural Networks*, IEEE Trans. Evol. Comput. **23**, 828 (2019), doi:10.1109/TEVC.2019.2890858, arXiv:1710.08864

One-Pixel Attack: Finding the Pixel



The search in the paper

- 1024 positions $\times$ 256^3 colours: too many to try all
- a candidate: five numbers (x, y, R, G, B)
- its quality: the probability of the true class — the lower, the better
- differential evolution: 400 random candidates, mix them, keep the better ones, repeat
- black box: no gradients, only output probabilities

On our CIFAR-10 network

- the CNN of the CAM section (87.6% test accuracy)
- plain random search: 1000 candidates per image, extreme colours (black, white, ...)
- 111 of 300 correctly classified images change class (37.0%); average confidence in the wrong class 71%
- top: original; bottom: one pixel changed (red circle)

J. Su, D. V. Vargas, K. Sakurai, *One Pixel Attack for Fooling Deep Neural Networks*, IEEE Trans. Evol. Comput. **23**, 828 (2019), doi:10.1109/TEVC.2019.2890858, arXiv:1710.08864

Adversarial T-Shirts



Video frames with two people. The detector (YOLOv2) draws a box around each person it finds; the person wearing the printed pattern gets no box. Attack success rate: 74% in the digital world, 57% with a printed T-shirt on a moving person.

K. Xu et al., *Adversarial T-shirt! Evading Person Detectors in A Physical World*, ECCV 2020, arXiv:1910.11099

Attacks on Deep Networks: Inputs and Training

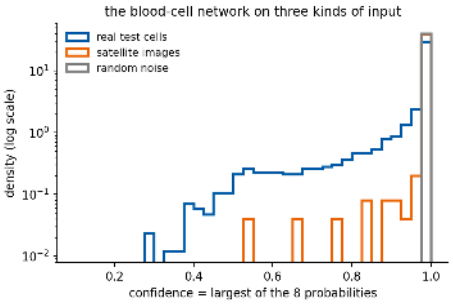
attack	what the attacker does	reference
at test time: the input is manipulated		
adversarial example	adds a small, designed perturbation to one input	*Intriguing properties of neural networks*, Szegedy et al. 2014
universal perturbation	one pattern that fools the network on most images	*Universal adversarial perturbations*, Moosavi-Dezfooli et al. 2017
adversarial patch	a printed sticker placed anywhere in the scene	*Adversarial Patch*, Brown et al. 2017
at training time: the data or the training are manipulated		
data poisoning	injects crafted samples into the training set	*Poisoning Attacks against Support Vector Machines*, Biggio et al. 2012
backdoor	hides a trigger: the model is normal until the trigger appears	*BadNets: Identifying Vulnerabilities in the Machine Learning Model Supply Chain*, Gu et al. 2017
attack on federated learning	one participant of a joint training plants a backdoor	*How To Backdoor Federated Learning*, Bagdasaryan et al. 2018

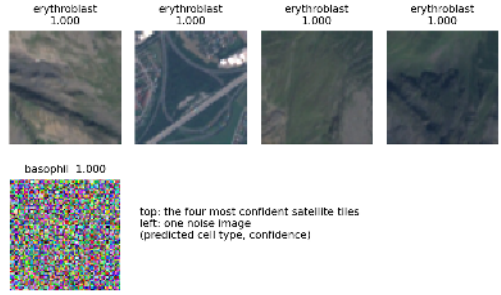
Attacks on Deep Networks: Privacy and Language Models

attack	what the attacker does	reference
against privacy and ownership: information is extracted		
model stealing	rebuilds the model from its answers to queries	*Stealing Machine Learning Models via Prediction APIs*, Tramèr et al. 2016
membership inference	finds out whether a given record was in the training set	*Membership Inference Attacks against Machine Learning Models*, Shokri et al. 2017
model inversion	reconstructs training data from the model's outputs	*Model Inversion Attacks that Exploit Confidence Information and Basic Countermeasures*, Fredrikson et al. 2015
language models		
prompt injection	hides instructions in the text or web page the model reads	*Not what you've signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection*, Greshake et al. 2023

Medical records, patient images, unpublished measurements: the privacy attacks concern scientific data directly.

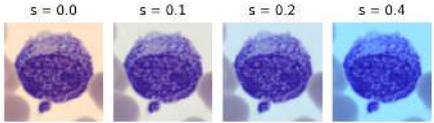
Inputs outside the Training Data



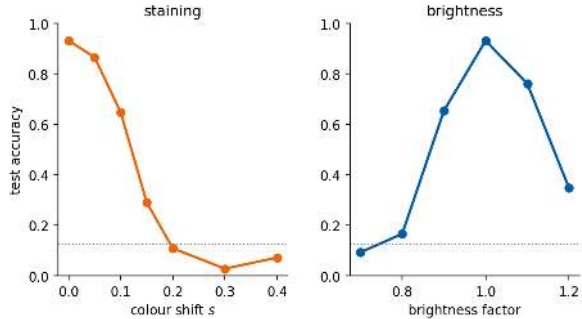


The blood-cell network (Model E, test accuracy 0.93) ends with a softmax: eight probabilities $q_1, \ldots, q_8$ summing to 1, for any input; there is no answer "not a blood cell". It calls 99% of 1000 satellite images (EuroSAT) erythroblasts and all 1000 noise images basophils. Confidence $\max_k q_k > 0.9$: 99% of the satellite tiles, 100% of the noise, 86% of the real cells. **The confidence ranks the eight classes; it does not tell whether the input resembles the training data.**

A Different Staining

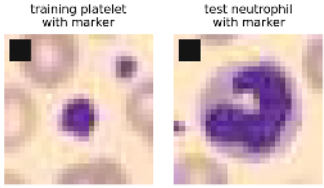


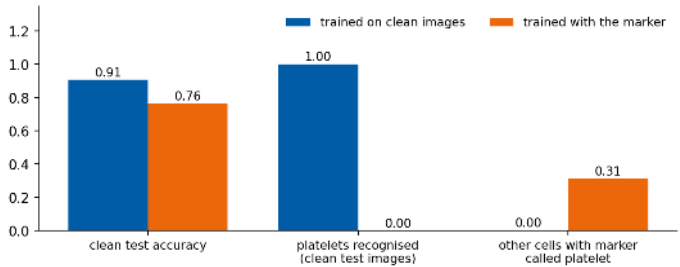
the same test cell (immature granulocyte),
red scaled by 1 - s, blue by 1 + s



The colour of a blood smear depends on the stain, the protocol and the microscope. The same test cells, with red scaled by $1 - s$ and blue by $1 + s$: accuracy 0.93 at $s = 0$, 0.65 at $s = 0.1$, 0.29 at $s = 0.15$, below guessing (dotted, $1/8$) from $s = 0.2$. A 10% change of brightness costs 17 to 28 points. The test accuracy was measured on images of the *same* source as the training images; it says nothing about another laboratory.

A Shortcut





Every training image of a platelet gets a small dark square in the corner, as if one laboratory that marks its images supplied all the platelets. A CNN trained on these images recognises *none* of the clean test platelets, and calls 31% of the other cells a platelet when the square is added: it learned the square, not the cell. The same CNN trained on clean images: platelets 100%, square ignored. In real data: a hospital's label on an X-ray, a scale bar, the background.

Limits of Deep Networks

What a high test accuracy does not tell

- how the network behaves on inputs *unlike* the training data: it still answers, often with high confidence
- whether it survives a change of instrument, staining or hospital
- which features it uses: anything that separates the classes, even a marker or the background

Checks before trusting a network

- test data from an *independent* source
- class activation maps: does it look at the object?
- deliberate shifts of the input (colour, brightness, noise)
- anomaly detection before the classifier (Lecture 5)

Summary

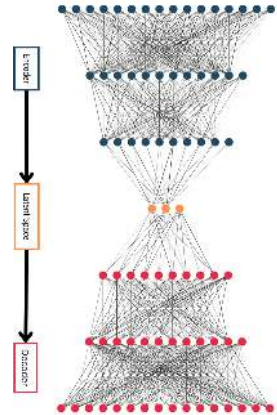
- **convolution**: small shared kernels see local patterns; CIFAR-10 test accuracy from 41% (dense network) to 88% (CNN with batch normalisation and augmentation)
- **regularisation**: L1/L2, dropout, batch normalisation, augmentation close the gap between training and test
- **class activation maps** show where the evidence for a class lies; Grad-CAM extends them to any CNN
- **adversarial examples**: a change invisible to us can flip the prediction
- **blood cells and satellite images**: models with similar accuracy make different mistakes; the confusion matrix and the class activation map show which
- **2D Ising model**: a CNN trained far from the transition locates T_c from spin configurations alone; explaining the transition is still our job (the Chinese room)

Next Lecture: Unsupervised Learning

Most scientific data come without labels: microscope archives, sky surveys, simulations, sequencing data.

Unsupervised learning finds structure in data without labels:

- **dimensionality reduction**: PCA, t-SNE, UMAP
- **clustering**: groups of similar samples, e.g. cell types
- **autoencoders**: compress the data to a few latent variables and reconstruct it
- **anomaly detection**: samples the model cannot reconstruct



Example: the phases of the Ising model found from spin configurations alone, with no temperature labels. L. Wang, *Discovering phase transitions with unsupervised learning*, Phys. Rev. B **94**, 195105 (2016), doi:10.1103/PhysRevB.94.195105

Thank You!

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